

WASP-50b

R-1685

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_131215 · created 2026-08-02T12:24:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456641.7343 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.017
Transit depth (Rp/Rs)^2	1.86 +/- 0.47 [%]
Orbital Inclination (inc)	85.42 +/- 1.68
Airmass coefficient 1 (a1)	1.224 +/- 0.015
Airmass coefficient 2 (a2)	-0.125 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	None
Optimal Aperture	3.57
Optimal Annulus	11.99
Transit Duration (day)	0.076 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.137 ± 0.017 vs 0.1404 (deviation 0.2σ)
Duration fit vs published	0.076 d vs 0.0754 d (deviation 0.04σ)
Mid-time O–C	0.2 min
Depth SNR	8.1
Residual scatter	1.24 %

Light curve

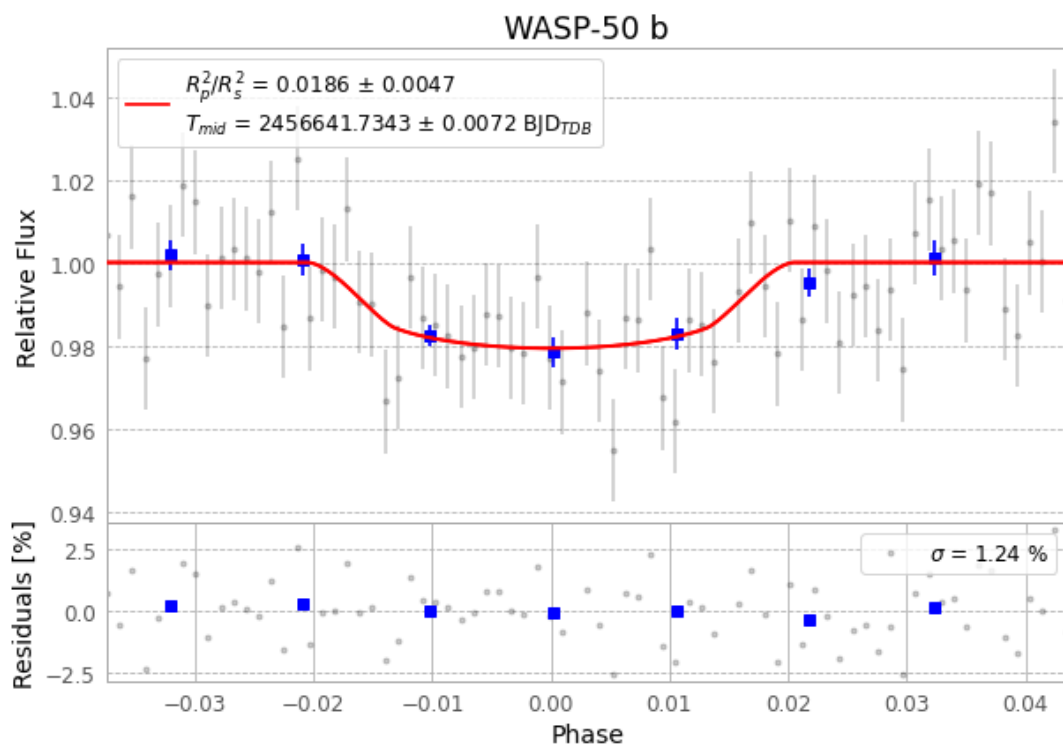


Figure 1 — FinalLightCurve_WASP-50 b_2013-12-14.png (SHA-256 795f2ecd9737f8f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a6cec98946813845...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50131215034513.FITS → WASP-50131215073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-131215.log (d2fdf43e5d66...), FinalLightCurve_WASP-50 b_2013-12-14.png (795f2ecd9737...), FinalLightCurve_WASP-50 b_2013-12-14.csv (feacbcd34ba6...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 12:24Z.